IMPROVISATIONAL ASPECTS OF IMAGE AND GESTURE SONIFICATION

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ABSTRACT

This paper/demonstration presents various approaches to gesture and image sonification and details those taken by the author in the creation of *SIHyperI* (Sonification and Intermedia Hyper-Instrument), a tool for interactive installations and real-time audio and visual performance. It discusses the aesthetic and functional issues that arise from the desire to use sonification for artistic and compositional purposes rather than for its more common application in auditory display. Interactivity, improvisation and intuitive parameter mapping are addressed along side their practical application.

1. INTRODUCTION AND MOTIVATION

The mere existence of the genres of opera, theater, and film attest to the long held desire for multimodal artistic presentations. They allow the exploration of relationships between artistic idioms that may reinforce each other or contrast in meaningful ways. Sonification, the conversion of data to abstract sounds, affords many novel relationships between images and music that can be exploited for this purpose. In the field of auditory display, which deals with the more practical aspects of representing large amounts of data as sound, these relationships are governed more by issues of perceptibility and clarity than aesthetic concerns, though they are not by any means mutually exclusive. However, the resulting audio is often more deterministic, utilitarian, and emotionless than what concertgoers are accustomed to. This paper details strategies for making sonifications more musical and expressive, as well as strategies for bridging the gap between still (timeless) images and the temporal nature of music.

2. SONIFICATION FOR COMPOSITION VS. AUDITORY DISPLAY

In their article *Listening to the Mind Listening: An Analysis of Sonification Reviews, Designs and Correspondences* [1] the authors discuss the results of a sonification competition with the criteria that submissions be both "data driven" and "musically satisfying". They note that many of the composers/submitters viewed these as somewhat contradictory objectives, and that one seemed to constrain the other. Indeed it seems hardly likely that a systematic parsing and mapping of data to sound will have the ebb, flow, and intuitive nuances that we are accustomed to in music. One solution to this dilemma is the introduction of interactivity. Even within the realm of auditory display, many authors have stressed the need for interactivity in the meaningful exploration of data [2,3]. Interactivity becomes even more important in musical/compositional applications where one is charged with the task of creating a musical composition, which usually implies a design and structure to the flow of stimuli over the duration of a piece such that it becomes musically satisfying. Kirsty Beilharz has documented her approach to using biologically inspired generative algorithms towards this aim [4].

2.1. SIHyperI

In *SIHyperI* interactivity is implemented by the use of a mouse to control the precise point or points (rows and columns) of the image to be sonified. Depending on the exact mappings chosen, directing the mouse towards parts of the image that are lighter or darker will impart various kinds of contrasts; contrast of pitch (in the case of 'notes'), frequency range (in the case of noise filtration or granular synthesis), pitch density, duration, timbre, timbral density, amplitude, or scale (diatonic, octatonic etc.). Such contrast, if used deliberately to form patterns, are of course the basic building blocks of musical structure, and thus given an image with a reasonable amount of contrast moving the mouse from one place to another on that image (in some perceptible pattern) will impart structure. Doing so in a manner rehearsed and refined can in turn impart nuance. Although *SIHyperI* allows one to select a chaotic algorithm for moving around the image, or to scan it left to right or top to bottom, experience shows that these are often perceived as compositional mistakes. By interacting/improvising with the mouse and feeling where in the image the mouse 'needs to go', a sense of drama can be created that mere scanning or relying on algorithms ignore. This is easily demonstrated using the software and becomes especially apparent when mapping brightness to loudness and/or density (notes/events per second) where the operator has an especially good control over the dramatic curve of the sonification over time. For interactive installations *SIHyperI* allows one to use the xy motion tracking from a web cam instead of a

mouse such that a participant merely points to the part of the image he/she wishes to sonify.

Other means to achieve a more traditionally musical sonification involve attention to texture. In sonification for purely auditory display, the texture often remains static and is quite often either strictly monophonic or pointalistic. A Geiger counter is a simple example of this. However, mapping the data to multiple sound generators, computationally affordable on contemporary computers, can result in more complex textures as granular synthesis algorithms may create a background while MIDI and samplers may provide a foreground, even if driven by the exact same data. Pitch/frequency choice also becomes an issue here, especially if one uses MIDI synthesizers or other equal tempered sound generators. Mapping the data coming from an image of a distant galaxy may lend itself to chromatic pitch mappings where as images of flowers may be complimented more by diatonic or pentatonic mappings, though choices will vary with taste. Again, this is where composition and auditory display impart different demands.

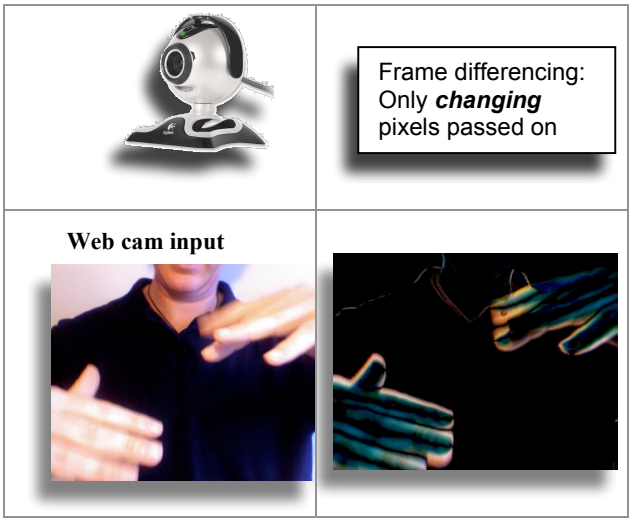


Figure 1. Frame differencing

3. IMAGE ANALYSIS OF GESTURES

Fortunately, for musical purposes, gestures exist in time and are thus relatively easy to capture and map when compared to still images. In *SIHyperI* the primary process used in gesture capture is frame differencing via a web cam; an absolute difference is calculated between the current and the last frame resulting in only delta pixels (pixels whose values have changed) passing through. Pixel values that have not changed from one frame to the next are sent as '0' (black) values (see figure 1). *SIHyperI* then tracks motion along the x and y-axes. Although there are many approaches to motion tracking, the method we use is somewhat unorthodox. 3 equally spaced columns from both the left and right side of the incoming matrix are extracted and their values averaged together to form a list of 127 values ranging from 0-255 (8 bit processing). To map these to MIDI notes the pixels with the highest values are found (post frame differencing, thus corresponding to the pixel that has changed the most) and it's position in the list calculated. Numbers at the start of the list correspond to pixels from the bottom of the matrix, which are mapped to lower notes, and numbers at the end of the list correspond to pixels at the top and are mapped to higher notes. The entire list is also used as an equalization curve for filtered noise (see figure 2).

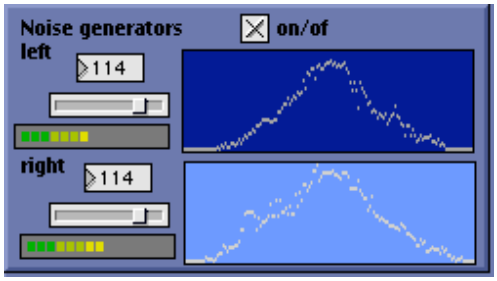


Figure 2. Equalization curves

The use of frame differencing has an important side effect for x/y axis gesture tracking in that the brightest pixel will still be lighter or darker depending on the amount of motion that generated it. Thus aside from its position it also has a *quality*, another dimension of information that can be used for expressive purposes. Similarly one very intuitive mapping is the total amount of motion generated to the amplitude of audio processing. This mapping is easily accomplished by summing (post frame differencing) the entire matrix. Thus high rates of movement correspond to high amplitudes and if there is no movement at all, there is no sound. This creates a very intuitive improvisational environment in which to create music with a web cam.

4. STILL IMAGE ANALYSIS

Still image analysis for use in the time domain is somewhat more challenging. One must first ask exactly what part of the image is to be analyzed and when. One approach is to analyze the entire image at once, allowing the mapping each of the pixel values to the frequency and amplitude of a tone generator, the transposition level of a granular synthesis instrument etc. Although this would enter the time domain for moving images and gestures, its application to still images simply creates a drone, which can be potentially annoying after relatively short periods. As stated above, *SIHyperI* uses the interactivity of the mouse to move around the image via crosshairs. One may select either the horizontal rows and/or vertical columns as data to sonify, or the precise point/pixel where the crosshairs meet. The latter then creates a 4-element set of data for the ARGB values of that pixel or, as is often the most intuitive, brightness, which is the weighted sum of the RGB values: Y = (0.299 * R) + (0.587 * G) + (0.114 * B). The image analysis outputs, for the whole image, a column, row, or an individual pixel, its redness, greenness, blueness, maximum, minimum or overall brightness, saturation,

contrast, dominant color and xyz axis position. Although analysis of other image characteristics are possible, their effectiveness is directly proportional to their perceptibility, and perceptibility is a key issue noted by Weinberg [5] and others.

5. MAPPING DATA TO DIFFERENT LAYERS/TEXTURES

Attention to texture is what often delineates a musically satisfying sonification (or composition for that matter) from a more one-dimensional aural display. As with an orchestra, this usually involves using more than one instrument, even if they share the same "data" as in a harmonic progression or scale. *SIHyperI* currently uses 9 sound generators of different kinds. Sound generators include the following with their detailed mappings.

5.1. MIDI

Finding individual MIDI notes is especially intuitive when tracking motion on the x, y, or z-axis. *SIHyperI* further splits the y-axis into two halves such that a performer or participant (in the case of an installation) can use their right and left hands to create counterpoint. In order not to overwhelm the instrument with data when sonifying an entire row or column of pixels, the values are placed in a table and accessed according to the number of notes per second chosen. This value seems most intuitive when mapped to brightness, accomplished, as are most mappings, by pull-down menus (see example 3). It is in this module that the use of scales, mentioned above, comes into play. In the case of the chromatic incoming values from 0-127 are simply matched directly to MIDI values. For other scales the incoming values are used as indexes for look-up tables containing the various scales.

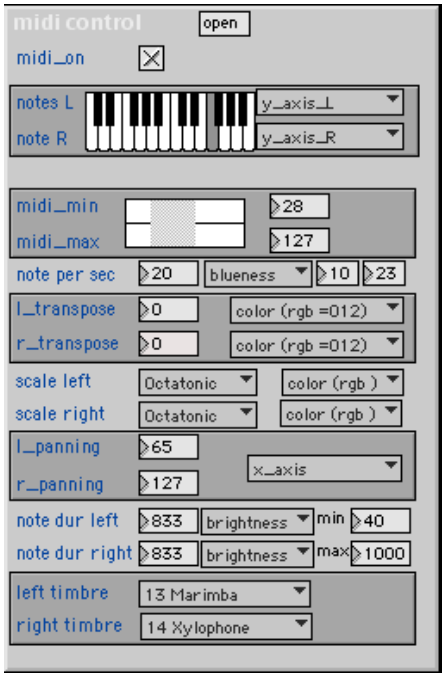


Figure 3. MIDI settings

5.2. Samplers

The same MIDI notes, along with their duration, panning and 'velocity' (amplitude) values are sent to two samplers. These feature, amongst other things, a user selectable number of notes per octave; with 24 notes/octave and a chromatic scale selected MIDI note values 60-84 would result in a one-octave quarter-tone scale. With 12 notes per octave they will simply double the MIDI notes (or substitute if MIDI is turned off) creating additional timbral richness. However, with more notes per octave the resultant pitches will move in similar (but not parallel) motion, creating counterpoint, and adding depth.

5.3. Noise generators

The noise generators (figure 2) use an algorithm by Tom Mays which takes the value for each pixel in a column, and uses that as a value for an equalization curve to filter a band of pink noise. Thus if the image is bright on top and dark below, the resultant audio with consist of mainly high frequencies and fewer low frequencies.

5.4. Granular synthesis

The nature of granular synthesis [6], with all of its various parameters, makes it a natural for utilizing various data coming from an image. In particular grain frequency, amplitude, density, and size are all parameters that respond intuitively to image analysis data, especially brightness and xyz position. *SIHyperI* employs two granular synthesis generators. The first being a cloud generator in which each of the grains are heard more or less discretely (see figure 4).

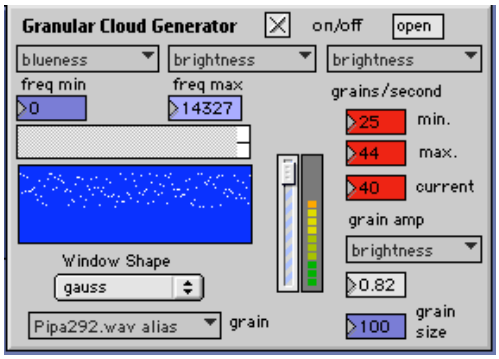


Figure 4. Granular Cloud Generator

The second is a sample granulizer which produces a constant sound as the grains transverse different parts of a sampled sound. In the case of the cloud generator the data is used to indicate the minimum and maximum frequency the cloud will use, where in the granulizer a downsized image produces 32 transposition values for each of the grains used (see figure 5).

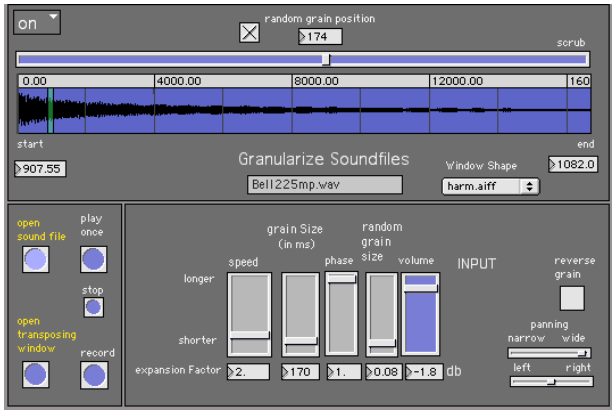


Figure 5. Sample granulator

5.5. Sound scrubber

SIHyperI uses a sound scrubber which is particularly useful when following the xyz motion tracking.

5.6. Sine wave modules

Lastly the program uses two-sine tone generating sound modules. The first uses a bank of 80 oscillators (additive synthesis), each tuned to frequencies determined by the relative brightness values of pixels (in a downsized matrix) of the entire image, or a column/row of a full image. The second, a phase synthesizer, uses values from the image analysis for frequency and amplitude values of two modulating oscillators, that then drive the phase values of two more oscillators in typical phase modulation synthesis.

6. CONCLUSION

Combinations of these 9 sound generators can be combined at will to form complex textures and timbres that sound far more musically interesting than any particular module on its own. This is key, as the motivation is to move beyond mere auditory display towards something more musically satisfying, which usually entails (amongst other things) complexities of timbre, attention to texture, and structure. These latter qualities can be achieved by real-time control, via the mouse, to allow the user to make expressive gestures and to plan structural movements through various contrasting areas of the image being sonified, thus leading to composition, as apposed to the often uninteresting parsing of data. *SIHyperI* can be downloaded (free) from: http://www.hkbu.edu.hk/~lamer/download.htm

7. REFERENCES

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